Spaces & closure

Vector spaces are sets in which addition and scalar multiplication obey familiar algebraic rules. This chapter develops the subspace criterion, explains how to combine subspaces.

1 Vector spaces

Definition 1.1 Vector space
A vector space over F is a set V equipped with addition and scalar multiplication satisfying the rules below. In these notes, F is R or C. Elements of V are vectors; elements of F are scalars.

Addition is associative and commutative. There is a zero vector 0, and every vector v has an additive inverse −v. Scalar multiplication satisfies

$$a(bv) = (ab)v, \qquad 1v = v,$$

$$a(u + v) = au + av, \qquad (a + b)v = av + bv.$$

The operations must also stay inside V.

Example 1.2 Vectors can be functions

For functions on a fixed domain, define addition and scalar multiplication pointwise:

$$(f + g)(x) = f(x) + g(x), \qquad (af)(x) = af(x).$$

Polynomials, continuous functions and ordinary coordinate vectors all provide examples of vector spaces, with the appropriate domain and scalar field.

Notation. The scalar 0 and the zero vector are different objects. In a function space, the zero vector is the function whose value is zero everywhere.

2 The subspace criterion

Definition 1.3 Subspace
A subset U of V is a subspace if it is a vector space with the addition and scalar multiplication inherited from V.

Theorem 1.4 Conditions for a subspace
A subset U of V is a subspace if and only if:
1. 0 belongs to U.
2. u + v belongs to U if u and v belong to U.
3. au belongs to U whenever a belongs to F and u belongs to U.

Proof. Necessity follows from the definition of a vector space. Conversely, suppose the three conditions hold. Zero belongs to U, and the two operations remain in U. Scaling by −1 supplies additive inverses. Associativity, commutativity and distributivity already hold in V, so they also hold for vectors in U. Thus U is a vector space.

Equivalently, check that U contains zero and that au+bv belongs to U for all u,v in U and all scalars a,b. This combines the two closure conditions into one calculation.

Example 1.5 A plane in R³

$$U = \{(x,y,z) \in R^3 : x - 2y + z = 0\}.$$

The zero vector satisfies the equation. If $u=(x_1,y_1,z_1)$ and $v=(x_2,y_2,z_2)$ belong to U, then the defining expression for au+bv is

$$a(x_1 - 2y_1 + z_1) + b(x_2 - 2y_2 + z_2) = 0.$$

Hence U is a subspace. Solving for x gives a second description of the same set:

$$(x,y,z) = y(2,1,0) + z(-1,0,1).$$

The equation tests whether a vector is allowed. The last expression constructs every allowed vector from two fixed vectors.

A contrasting example. Replacing the right side by 3 produces a set that misses zero. It is therefore not a subspace, even though it is still a plane.

3 Intersections unions and sums

Intersections preserve closure

If U and W are subspaces, zero belongs to both. For u,v in U∩W, every linear combination au+bv belongs to U and to W. It therefore belongs to the intersection. The same argument works for an arbitrary family of subspaces.

A union can fail closure

Let U be the x-axis and W the y-axis in R^2. Their union contains (1,0) and (0,1), but it does not contain (1,1). Thus U∪W fails closure under addition.

Definition 1.6 Sum of two subspaces
U + W = {u + w : u ∈ U and w ∈ W}.

Theorem 1.7 The smallest containing subspace
U+W is a subspace containing U and W. Moreover, every subspace containing both U and W also contains U+W.

Proof. Zero belongs to U+W. For $x=u_1+w_1$ and $y=u_2+w_2$, a linear combination has the form

$$ax + by = (au_1 + bu_2) + (aw_1 + bw_2).$$

The first parenthesis belongs to U, and the second belongs to W. Hence the sum is a subspace. It contains U because u=u+0, and it contains W because w=0+w.

Finally, let S be any subspace containing U and W. Since u,w belong to S, closure implies u+w belongs to S. Therefore U+W⊆S. This is the precise meaning of “smallest”: smallest by set inclusion.

Example 1.8 Two planes

In R^3, take U={(a,b,0)} and W={(0,c,d)}. Every vector (x,y,z)=(x,y,0)+(0,0,z) belongs to U+W, so U+W=R^3. Their intersection is the y-axis.

4 Direct sums

Definition 1.9 Direct sum
The sum U+W is direct if every vector in it has exactly one expression u+w with u∈U and w∈W. In this case we write U⊕W.

Theorem 1.10 Uniqueness criterion
For two subspaces U and W, the sum U+W is direct if and only if U∩W={0}.

Proof. First suppose that U + W is a direct sum. If v ∈ U ∩ W, then 0 = v + (−v), where v ∈ U and −v ∈ W. By the unique representation of 0 as the sum of a vector in U and a vector in W, we have v = 0. Thus U ∩ W = {0}, completing the proof in one direction.

To prove the other direction, now suppose U ∩ W = {0}. To prove that U + W is a direct sum, suppose u ∈ U, w ∈ W, and

$$0 = u + w.$$

To complete the proof, we only need to show that u = w = 0 . The equation above implies that u = −w ∈ W. Thus u ∈ U ∩ W. Hence u = 0, which by the equation above implies that w = 0, completing the proof.

Example 1.11 Even and odd functions

Let E and O denote the real vector spaces of even and odd functions on R. For any real-valued function f, define

$$e(x) = [f(x) + f(-x)]/2,$$

$$o(x) = [f(x) - f(-x)]/2.$$

Substituting −x shows that e is even and o is odd. Their sum equals f. If h is both even and odd, then h(x)=h(−x)=−h(x), so h(x)=0 for every x. Hence E∩O={0}, and the decomposition is unique.

Remark 1.12 Three summands
Pairwise zero intersections do not guarantee a direct sum of three spaces. The x-axis, y-axis and line y=x in R^2 have pairwise zero intersections, but
(1,1) = (1,0)+(0,1)+0 = 0+0+(1,1).

5 Further discussion

When is a union a subspace

Proposition 1.13
For two subspaces U and W, $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Proof. If one space contains the other, the union equals the larger subspace. Conversely, assume neither inclusion holds. Choose $u \in U$ with $u \notin W$, and $w \in W$ with $w \notin U$. If the union were a subspace, it would contain $u+w$.

If $u+w$ belongs to U, subtracting u gives $w \in U$, a contradiction. If $u+w$ belongs to W, subtracting w gives $u \in W$, again a contradiction. Thus the union cannot be a subspace.

Reading: Axler, Linear Algebra Done Right, fourth edition, Sections 1A–1C.

6 Exercises

1. For which real c is $\{(x,y,z) : x+y-z=c\}$ a subspace of R^3?

2. Let $U=\{(a,b,0)\}$ and $W=\{(0,c,d)\}$. Describe $U \cap W$ and explain why $U+W$ is not direct.

3. Prove that $\{p \in R[x] : p(1)=0\}$ is a subspace of the real polynomial space.

4. Classify $\{(x,y) \in R^2 : x^2+y^2=0\}$. Does the same conclusion hold over C?

5. Prove that the translate v_0+U contains the whole line through any two of its distinct points.